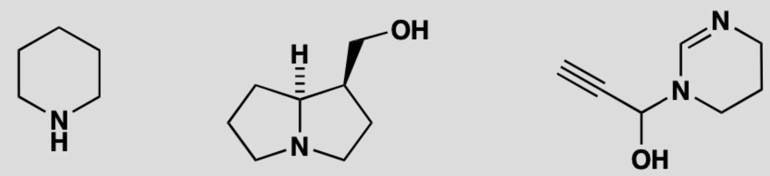
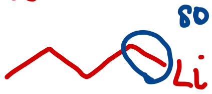
PROBLEM 4

What would you expect to be the site of (a) protonation and (b) deprotonation if these compounds were treated with the appropriate acid or base? In each case suggest a suitable acid or base and give the structure of the products.



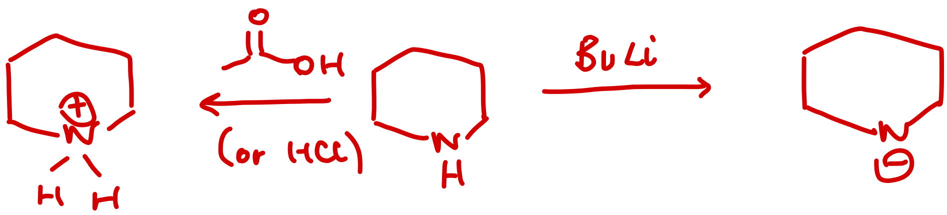
BuLi

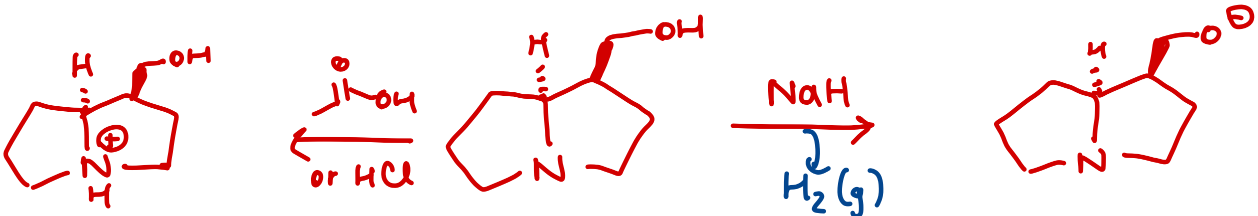


source of electrons

NaH = Na^+ H^-

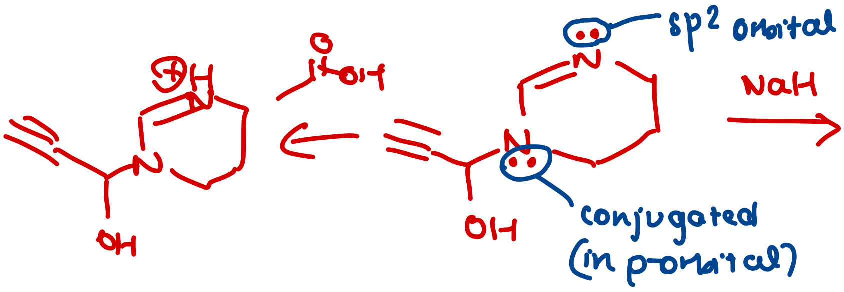
strong base

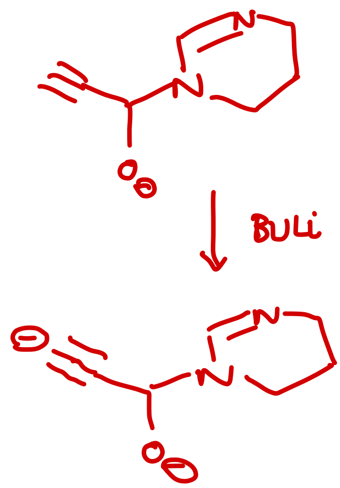




increase in entropy

$S(g) > S(l) > S(s)$

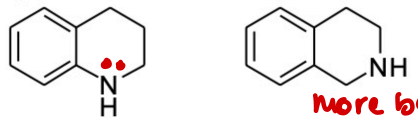




Alkyne >	Alkene >	Alkane
sp	sp^2	sp^3
$\boxed{50\%}$	33%	25%

2. Which of the following is the more basic?

a) delocalisation



more basic

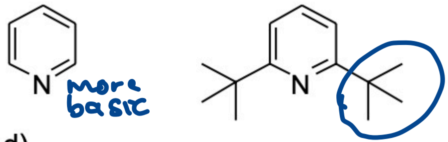
b) NaOEt NaSEt

more basic

S less electronegative

S 3p vs O 2p

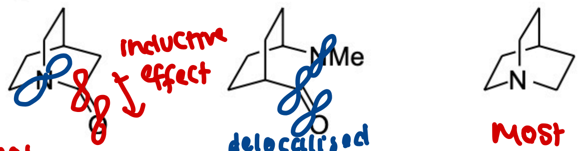
c)



more basic

steric hindrance

d)



inductive effect

not delocalised

delocalised

most basic

$$CH_3CO_2H \rightleftharpoons CH_3CO_2^- + H^+$$

4. a) Explain the significance of the following pK_a data for ethanoic acid (CH_3CO_2H) measured in a range of solvents at 298 K.

Solvent:	H_2O	MeOH	DMSO	MeCN
pKa:	4.8	9.0	12.5	23.0

less polar

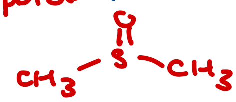
no H bonds

$$K_a = \frac{[H^+][CH_3CO_2^-]}{[CH_3CO_2H]}$$

H_2O polar protic

MeOH polar protic

DMSO



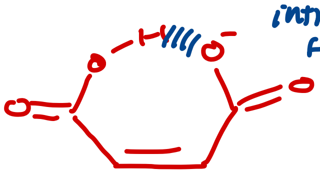
polar aprotic

MeCN polar aprotic

b) Explain why maleic acid has a lower pK_{a1} (first dissociation) then fumaric acid but a higher pK_{a2} (second dissociation)

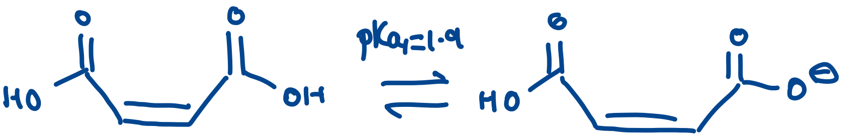
maleic acid
pK_{a1} = 1.9
pK_{a2} = 6.2

fumaric acid
pK_{a1} = 3.0
pK_{a2} = 4.4



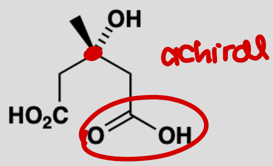
intramolecular H-bond

needs to be broken in pK_{a2}



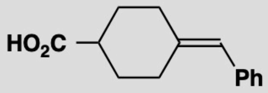
PROBLEM 1

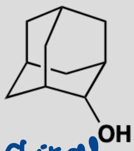
Are these molecules chiral? Draw diagrams to justify your answer.



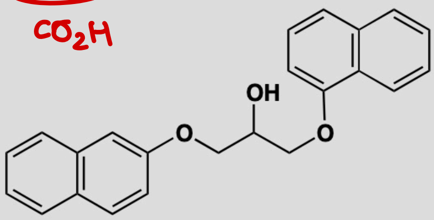
achiral

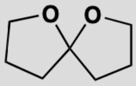
CO_2H



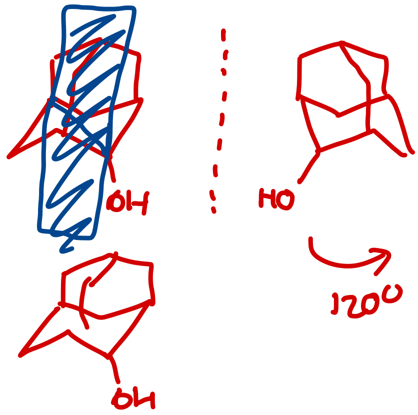


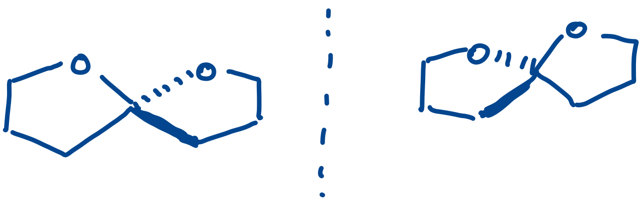
achiral





chiral





non-superimposable